

Enterprise RAG Chatbot University of the Cumberlands

Final Project Report
Ethics in Artificial Intelligence

Team Members

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1. System Architecture

The system operates as two pipelines sharing a common vector store. The **ingestion pipeline** (offline) scrapes 2,092 pages from ucmerlands.edu, converts them to markdown, chunks by section headings, embeds with BAAI/bge-base-en-v1.5 (768-dim), and indexes into ChromaDB (20,520 chunks, cosine HNSW). The **query pipeline** (real-time) accepts user questions via a chat widget, expands short queries with conversation context, embeds the query, retrieves top-15 chunks, passes top-10 to AWS Bedrock Claude Sonnet for generation, and returns a cited response.

Pipeline flow: Scrape → Markdown Conversion → Section-Aware Chunking → BGE Embedding → ChromaDB → Cosine Retrieval → Claude Sonnet Generation → Chat Response with Citations. FastAPI orchestrates all components; session history (3 turns) enables conversational follow-ups.

2. Retrieval & Generation Design

Embedding model: BAAI/bge-base-en-v1.5 — selected over all-MiniLM-L6 (lower MTEB score) and OpenAI text-embedding-3-small (50 min API latency vs 4 min local). Runs on Apple Silicon MPS GPU. **Generation LLM:** AWS Bedrock Claude Sonnet — chosen for strong instruction-following, 200K context window, and managed infrastructure (~3.7s avg latency).

System prompt design: Six rules enforce grounding: (1) answer only from context, (2) 2-4 paragraphs max, (3) include clickable [text](URL) citations, (4) refuse gracefully when context is insufficient, (5) professional tone with no emojis, (6) never reference "context" or "provided information." Citations are enforced by presenting each chunk with [Source N: Title | URL] metadata.

Insufficient context handling: Two-layer detection — retrieval threshold of 0.3 cosine similarity triggers a pre-written refusal without calling the LLM; additionally, Rule 4 instructs the model to acknowledge gaps rather than fabricate answers.

3. Chunking & Re-Chunk Experiment

We compared **fixed-size chunking** (500 chars, 50-char overlap) against **section-aware chunking** (splits at ## headings, sentence-boundary aware, 50-1500 char range, context prefix). Section-aware chunking improved P@5 from 72.3% to 89.8% (+17.5pp) and accuracy from 61.2% to 73.6% (+12.4pp).

Metric	Fixed-Size	Section-Aware	Delta
P@5	72.3%	89.8%	+17.5pp
R@10	79.1%	88.7%	+9.6pp
Accuracy	61.2%	73.6%	+12.4pp
Chunks	18,400	20,520	+11.5%

Chunk ID scheme: {url_hash}_{chunk_index} (SHA-256 truncated). **Re-chunk process:** Dropped the entire ChromaDB collection, re-chunked from cached markdown (no re-crawl), re-embedded all 20,520 chunks, verified zero stale data by confirming count match and spot-checking that all returned chunks use the new heading-prefix format.

4. Version Control & Team Collaboration

Repository: github.com/rmakala38838/uc-rag-chatbot. Feature-branch workflow with PR reviews required before merge to main. Branches named feature/{member}-{task}. Sensitive files (.env) excluded via .gitignore.

Member	Role	Contributions
Mounica Dayana	Product Manager	Proposal, requirements, evaluation criteria, presentation
Eswari Ankitha Datla	Backend Dev	FastAPI, ChromaDB integration, Bedrock client, query expansion
Rajesh Makala	Frontend Dev	Chat widget, landing page, UC branding, README, deployment
Sandeep Bagam	Data Scientist	Scraper, chunking, embeddings, evaluation scripts, metrics

5. Ethics Considerations

Risk	Mitigation
Hallucination	Context-only prompt; 0.3 threshold refusal; 100% faithfulness in eval
Outdated info	Source URLs in every response; timestamps in metadata; re-scrape pipeline
Over-reliance	Disclaimer footer; explicit refusal on low confidence
Crawl ethics	100ms delay; robots.txt; public pages only; academic use
Privacy	No query logging; session-only memory; no PII collected

Limitations: OOD refusal imperfect (2/5 false accepts); no automated re-scrape schedule; English only; no user feedback mechanism; embedding bias unmeasured. **With more time:** thumbs-up/down feedback, confidence scores shown to users, formal bias audit across 165+ programs, multilingual support.

6. Challenges & Solutions

Challenge 1 — Embedding speed: OpenAI API took 50 minutes for 20K chunks. **Solution:** Switched to local BGE-base on MPS GPU — 4 minutes (12.5x faster), no API cost, same quality (MTEB 63.55 vs 62.30).

Challenge 2 — Follow-up failures: Short queries ("tell me more") retrieved irrelevant chunks. **Solution:** Query expansion detects short/pronoun-heavy queries and prepends topic from chat history (e.g., "MSIT program — tell me more").

Challenge 3 — OOD detection: System answered all off-topic questions (0% refusal). **Solution:** Two-layer refusal: cosine threshold (0.3) + prompt instruction. Improved to 90% refusal rate.

7. Evaluation

50 golden questions with ground-truth answers from UC documents. Two methods: keyword-based metrics and LLM-as-Judge (Claude Opus evaluates correctness, completeness, faithfulness, relevance, citation quality).

Headline Metrics (with 95% CI, Wilson score, n=50)

Method	Metric	Score	95% CI
Keyword	Precision@5	89.8%	±4.2%
Keyword	Recall@10	88.7%	±4.5%
Keyword	Accuracy	73.6%	±6.1%
Keyword	Faithfulness	100%	±0%
LLM Judge	Correctness	80.8%	±5.5%
LLM Judge	Completeness	76.4%	±5.9%
LLM Judge	Faithfulness	82.8%	±5.2%
LLM Judge	Relevance	87.2%	±4.6%
LLM Judge	Citation Quality	84.0%	±5.1%

Statistical significance: McNemar's test comparing section-aware vs fixed-size chunking accuracy: $\chi^2=8.33$, $p=0.0039$ (significant at $\alpha=0.01$). Spearman correlation between keyword and LLM-Judge scores: $\rho=0.72$, $p<0.001$.

Error Analysis — 15 Hand-Labeled Failures

Failure Mode	Count	%	Example & Root Cause
Incomplete retrieval	6	40%	Multi-chunk answers; only partial content in top-10
False accept (OOD)	2	13%	Off-topic query matched tangential UC content
Wrong source	2	13%	General page retrieved instead of specific section
Missing content	2	13%	Info in footer/sidebar not captured by chunker
Vague/generic	2	13%	Correct but lacks specific details from source
Verbose/redundant	1	7%	Repeated same fact from multiple retrieved chunks

Dominant failure (40%): answers spanning multiple chunks exceed top-K retrieval capacity.

8. Future Work

- 1. Parent-document retrieval:** Store small chunks for precise matching but return the full parent section to the LLM — directly addresses the 40% incomplete-retrieval failure mode.
- 2. Hybrid BM25 + dense search:** Add sparse keyword retrieval with Reciprocal Rank Fusion for better recall on course codes, dollar figures, and proper nouns that embeddings handle poorly.
- 3. Automated freshness pipeline:** Weekly cron scrape with content-hash comparison; incremental re-chunk of changed pages only; atomic ChromaDB collection swap.
- 4. User feedback loop:** Thumbs-up/down on responses; negative feedback flags questions for human review and expands the golden test set over time.

References

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